

Yuan Tuo

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Education

2020–26	Ph.D., Economics, University of California, Davis (expected)
2018–19	B.Ec., Economics (First Class Honours), The University of Sydney
2017	Exchange, University of Wisconsin–Madison
2015–18	B.Ec., Economics, The Australian National University
2014–15	Wuhan University, China

Research Interest

Econometrics, Applied Econometrics

Working Papers

Optimal Treatment Under Spillover Effects and Noncompliance (Job Market Paper)

Abstract: This paper considers optimal treatment allocation in the presence of spillover effects in both compliance and outcome. We show that spillovers on an individual's outcome can be from unconnected individuals in the network when there are spillovers in compliance. For optimal treatment allocation, we empirically maximize welfare subject to some budget constraint or minimize cost subject to some welfare goal based on the characteristics of the target population. We propose an optimal randomized treatment rule and show that it achieves uniform asymptotic welfare efficiency with a $n^{-1/2}$ convergence rate and is uniformly asymptotically feasible.

Causal Inference in RCTs with Multiple Arms

Abstract: When there are more than one treatment group in an RCT with non-compliance, local average treatment effects (LATE) are not comparable as each treatment has different compliers. We provide a causal method to compare two treatments in the presence of non-compliance. In particular, we derive bounds on the average difference between pairs of treatment effects of those who would comply with both treatments, as well as bounds on the average treatment effects of other sub-populations.

Work in Progress

Inference of Marginal Treatment Effect (MTE) Estimator (with Takuya Ura)

References

Takuya Ura

UC Davis
Associate professor
takura@ucdavis.edu

Shu Shen

UC Davis
Associate professor
shushen@ucdavis.edu

Colin Cameron

UC Davis
Professor
accameron@ucdavis.edu

Presentations

- 2025 California Econometrics Conference
- 2025 Midwest Econometrics Group Conference

Fellowships and Awards

- 2023 Dean's Graduate Summer Support Award, University of California, Davis
- 2021 First-Year Fellowship, University of California, Davis

Teaching Experience

- 2020–25 **Teaching Assistant, University of California, Davis**
Undergraduate
Analysis of Economic Data, Econometrics, Economic & Financial Forecasting,
Economics & Business Data Analytics, Principles of Microeconomics,
Principles of Macroeconomics, Intermediate Macro Theory, Public Finance
Graduate (Ph.D. level)
Econometric Methods, Cross-Sectional Econometrics
- 2019 **Tutor, The University of Sydney**
Introductory Microeconomics, Introductory Macroeconomics

Referee Service

Econometrics Review

Software Skills

R, Stata, Matlab

Languages

Mandarin (native), English (fluent), Japanese (JLPT N1)